



OPEN Improving the explainability of autoencoder factors for commodities through forecast-based Shapley values

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Autoencoders are dimension reduction models in the field of machine learning which can be thought of as a neural network counterpart of principal components analysis (PCA). Due to their flexibility and good performance, autoencoders have been recently used for estimating nonlinear factor models in finance. The main weakness of autoencoders is that the results are less explainable than those obtained with the PCA. In this paper, we propose the adoption of the Shapley value to improve the explainability of autoencoders in the context of nonlinear factor models. In particular, we measure the relevance of nonlinear latent factors using a forecast-based Shapley value approach that measures each latent factor's contributions in determining the out-of-sample accuracy in factor-augmented models. Considering the interesting empirical instance of the commodity market, we identify the most relevant latent factors for each commodity based on their out-of-sample forecasting ability.

Keywords Explainability, Neural networks, Nonlinear factor models, Shapley value, Commodities

Machine learning is a well-developed theoretical framework that is a branch of the wide context of artificial intelligence. It aims to use data and algorithmic procedures to mimic the way humans learn.

Evidently, the versatility of this theoretical structure is the ground of the popularity of machine learning in all the fields of applied science. This paper enters this debate by facing a machine learning-related issue having remarkable applications in the field of finance, with a specific interest in the area of commodities. On this, we are close to the strand of literature dealing with machine learning techniques for commodities^{1,2}.

One of the key instruments used in machine learning is the neural network. Such a tool is a device imitating the human brain when processing flows of information—for us, signals. It is composed of neurons—or, simply, nodes—of different types. The input nodes are the nodes where the signal enters the network; the hidden nodes are devoted to the transformation and elaboration of the starting signal; the output nodes are associated with the final outcome of the transformed signal³.

In this environment, autoencoders are unsupervised learning instruments based on the reduction of the dimensionality of the considered models^{4,5}. On this, they can be seen as a nonlinear counterpart of the principal components analysis (PCA)⁶ associated with the factor models, that are widely exploited in the context of commodities^{7–13}. Compared to PCA, however, autoencoders capture complex relationships and nonlinearities while estimating latent factors. Therefore, it is not surprising that autoencoders have been used for estimating nonlinear factor models in finance^{14–16}.

Despite this relevance, the employment of autoencoders for commodities is quite neglected. To the best of our knowledge¹⁵, is the only paper applying autoencoder factor models to commodities.

This paper fills this gap. We deal with autoencoders when they are applied to factor models in the field of commodities. In this context, one of the major problems is to treat the explainability of autoencoders, which is lower than what one has for PCA¹⁷. Indeed, PCA components are easier to interpret because they are linear

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combinations of the original variables with clearly quantified variance. Autoencoders, on the other hand, use neural networks to learn complex, non-linear representations, making their encoding and decoding processes more opaque¹⁸. We here specifically face the issue of explainability by exploring the relevance of the nonlinear latent factors by applying a suitably defined notion of Shapley value, introduced by Lloyd Shapley in¹⁹ in game theory. The Shapley value is a well-recognized instrument for determining the importance of a player by measuring how the presence of the considered player affects the payoffs of the coalitions (see²⁰ and the recent handbook²¹).

The Shapley value is here applied to assess the importance of the latent factors for out-of-sample accuracy when making forecasts through factor-augmented models.

Our approach is in line with the contributions dealing with the application of the Shapley value in the environment of forecasting exercise^{22,23}. Furthermore, our methodological study aims to enhance the explainability of machine learning methods, building on discussions and outlines by^{24–27}. Specifically,²⁴ proposes a framework based on the Shapley–Taylor decomposition to perform parametric inference in machine learning models, offering insights into statistical learning processes and facilitating the generalization of experimental findings. Meanwhile,²⁵ introduces a global explainable artificial intelligence method using Lorenz decomposition, providing a unified variable importance criterion that combines predictive accuracy with explainability, illustrated through Bitcoin price prediction²⁶. Explores the application of Shapley values in random forest and support vector machine models for molecular machine learning, revealing differences in learning characteristics and providing chemically intuitive explanations for accurate predictions. Finally,²⁷ discusses the exploitation of the Shapley value concept for explaining support vector machine models in pharmaceutical research, highlighting the development of a computationally efficient approach by using radial basis function kernels. We take inspiration by the approach of^{22,23} but, differently from them and from the papers quoted above, we rank the factors identified by the autoencoders in terms of their importance in forecasting according to the Shapley value, thereby highlighting the explanatory power of each factor.

The theoretical proposal is validated through an empirical instance of the daily series of the returns of 22 commodities. The results are quite interesting, and offer room for further developments of the study. Indeed, for each commodity, we identify the key latent factors. Considering five factors, we find that the first latent factor is the most representative of energy-related commodities, whilst the third factor explains most agricultural commodities. Moreover, we find that the last factor mainly relates to metal commodities. Therefore, over five factors, only some of them are important for explaining the commodity market in out-of-sample terms. These considerations not only shed light on the information included in each latent factor, making the autoencoder more explainable, but they also represent a heuristic approach that can guide the selection of the number of relevant factors.

The rest of the paper is organized as follows. Section 2 describes the methodologies advanced in the paper, with a description of the factor models with autoencoders and the forecast-based Shapley value. Section 3 describes the dataset employed for the analysis. Section 4 outlines the results and provides a discussion of them. The last section offers some conclusions and illustrates lines for future research.

Methodology

In what follows we discuss the methodology adopted for the estimation of nonlinear factor models. Moreover, we present the procedure adopted to improve the explainability of factor models with autoencoders.

To provide a better intuition of the empirical instance proposed in the present paper, we will present the methodological framework in the context of commodities.

The proposed approach builds on the forecasting ability of latent factors. Autoregressive models are commonly used in time series modelling, but past returns alone may not capture all relevant information for forecasting commodity returns. Literature suggests using both observable and latent factors for better forecasts, leading to the preference for factor-augmented models²⁸. Observable factors include measurable variables such as interest rates, exchange rates, and others, whilst latent factors are not directly observable and represent underlying economic conditions or market sentiments. Including latent factors helps capture complex interactions that observable factors alone might miss. Therefore, we evaluate the forecasting ability of latent factors within the framework of factor-augmented linear regression models.

Factor models with autoencoders

The standard approach for estimating latent factors is by approximate factor models. Approximate factor models assume that observed time series stored in the matrix $\mathbf{X} = (x_{i,t}; i = 1, \dots, N; t = 1, \dots, T)$ can be decomposed into common factors $\mathbf{F}$ and idiosyncratic components $\mathbf{Z}$, that is,

$$\mathbf{X} = \mathbf{F}\mathbf{\Lambda}' + \mathbf{Z}, \quad (1)$$

where $\mathbf{F} = (f_{k,t}; k = 1, \dots, K; t = 1, \dots, T)$ with K the number of latent factors, $\mathbf{Z} = (z_{i,t}; i = 1, \dots, N; t = 1, \dots, T)$ and $\mathbf{\Lambda} = (\lambda_{i,k}; i = 1, \dots, N; k = 1, \dots, K)$. The principal component estimator²⁹, PCE, is commonly adopted to estimate the factors and the loadings from the covariance matrix of the observed time series. In particular, the PCE involves the following steps. First, the data is demeaned and the sample covariance matrix $\hat{\mathbf{S}}$ is computed. Then, the eigenvalues and eigenvectors of $\hat{\mathbf{S}}$ are obtained and ordered in decreasing order. The top K eigenvectors to form the loading matrix $\mathbf{\Lambda}$ are chosen and the common factors $\mathbf{F}$ are estimated using the selected eigenvectors. Although highly explainable, the PCE approach assumes linearity between factors and the commodity time series. The linearity assumption is inconvenient in the case of the commodity market, and the introduction of nonlinearities may be preferable. Nonlinear factor models can be estimated using autoencoders¹⁴.

An autoencoder is a distinctive type of neural network designed to have its outputs closely match the input variables. In this model, the input data passes through a limited number of neurons within the hidden layer(s),

creating a condensed version of the input (encoding). This compact representation is then expanded and translated back into the output layer (decoding). Since the model relies solely on the input data, it operates as an unsupervised learning mechanism⁴. Figure 1 provides an example of an autoencoder architecture. In this example, we have a single layer and two hidden nodes, which represent the factors. The first part of the figure, where the input data is compressed with the factors, is called the encoder. The second part, where the input is reconstructed from the factors, is called the decoder.

Generally speaking, an autoencoder is a neural network with L hidden layers. Let $K^{(l)}$ denote the number of neurons in the l -th layer, and define the output of neuron k in layer l as $x_k^{(l)}$. The vector of all outputs for this layer is denoted as $\mathbf{x}^{(l)} = (x_1^{(l)}, \dots, x_{K^{(l)}}^{(l)})'$. In each hidden layer, inputs from the previous layer are transformed according to a nonlinear activation function $g(\cdot)$ before being passed to the next layer. The recursive output formula for the neural network in layer $l > 0$ is then

$$\mathbf{x}^{(l)} = g(\mathbf{b}^{(l-1)} + \mathbf{W}^{(l-1)}\mathbf{x}^{(l-1)}) \quad (2)$$

where $\mathbf{W}^{(l-1)}$ is a $K^{(l)} \times K^{(l-1)}$ matrix of weight parameters, and $\mathbf{b}^{(l-1)}$ is a $K^{(l)} \times 1$ vector of bias parameters. Following^{14,15}, in this paper we use the rectified linear unit (ReLU) activation function. The number of parameters in each hidden layer l is $K^{(l)}(1 + K^{(l-1)})$. The final output of the autoencoder,

$$\hat{\mathbf{x}} = g'(\mathbf{b}^{(L)} + \mathbf{W}^{(L)}\mathbf{x}^{(L)}) \quad (3)$$

which shares the same dimension as the input, is employed to approximate the input itself.

The machine learning literature has long recognized the close connection between autoencoders and PCA^{4,30} and, by extension, between autoencoders and latent factor asset pricing models¹⁴. When the autoencoder has one hidden layer and a linear activation function, it is equivalent to the PCA estimator for linear factor models¹⁴.

Autoencoders, like neural networks more broadly, have many advantages relative to traditional linear factor models. In particular, the high capacity of a neural network model enhances its flexibility to construct the most informative features from data. The main weakness of autoencoders is that the results are less explainable than those obtained with the PCA. In this paper, we propose an approach to improve the explainability of autoencoders in the context of nonlinear factor models. In particular, we measure the relevance of nonlinear latent factors using a forecast-based Shapley value approach by measuring each latent factor's contributions in determining the out-of-sample loss.

Forecast-based Shapley value

Shapley values draw on coalitional game theory and build on the analogy between the predictors in a model and players in a cooperative game earning payoffs. In machine learning the payoff corresponds to the contribution

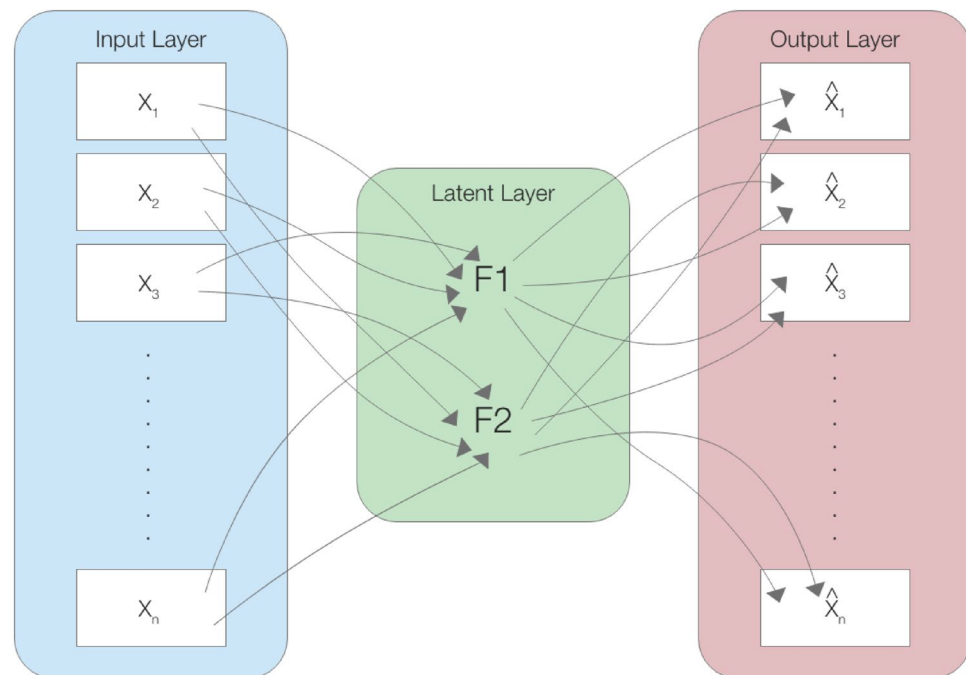


Figure 1. Example of autoencoder architecture for an input matrix $\mathbf{X}$. In this example, we have a single latent layer and two hidden nodes, which represent the factors. The first part of the figure is called the encoder, whilst the second is the decoder. The outputs are the same as the inputs.

of an individual variable to the model's prediction. Given its informative content, Shapley value is commonly used in the literature to improve the explainability of machine learning methods²⁵.

Let us define the notation adopted for the Shapley value definition in our context. Consider the following linear model²⁸

$$x_{i,t} = \alpha + \sum_{j=1}^p \beta_j x_{i,t-j} + \sum_{j=1}^p \gamma_j' \mathbf{f}_{t-j} + \varepsilon_{i,t} \quad (4)$$

where $x_{i,t}$ is the commodity i at time t , α is a constant term, p is the number of lags, β_j is the parameter associated with the j -th lag of $x_{i,t}$, γ_j is the vector of K regression parameters associated with the j -th lagged latent factors $\mathbf{f}_{t-j} = [f_{1,t-j}, \dots, f_{K,t-j}]'$. Therefore, we assume there exist K latent factors that are useful to predict the returns of a commodity $x_{i,t}$. Let us define by $S = \{1, \dots, K\}$ the set of these predictors. We note that we can use information criteria to select the number of lags p to be included in (4), for instance, see³¹. In the section or results we will discuss this point in details.

The Shapley value for factor k in a model trained using T observations is often given by

$$\phi_k(\mathbf{f}_t) = \sum_{Q \subseteq S \setminus \{k\}} \frac{|Q|!(K - |Q| - 1)!}{K!} [\xi_{Q \cup \{k\}}(\mathbf{f}_t) - \xi_Q(\mathbf{f}_t)] \quad (5)$$

for $k \in S$, where Q is a subset of factors (i.e., a coalition), $|Q|$ is the cardinality of Q , and $\frac{|Q|!(K - |Q| - 1)!}{K!}$ is a combinatorial weight. The dependence of the quantities in (5) on T is conveniently omitted. The expression $\xi_Q(\mathbf{f}_t)$ is the prediction of the fitted model conditional on the predictors in coalition Q , so that $\xi_{Q \cup \{k\}}(\mathbf{f}_t) - \xi_Q(\mathbf{f}_t)$ measures the difference in the value of the predictions, conditional on the factors included in the coalition Q , when the k -th factor is included in the conditioning information set. The difference $\xi_{Q \cup \{k\}}(\mathbf{f}_t) - \xi_Q(\mathbf{f}_t)$ is computed for all possible coalitions of $K - 1$ factors that exclude predictor k , with each quantity receiving the weight $\frac{|Q|!(K - |Q| - 1)!}{K!}$ in the summation. Differentials in terms of adjusted- R^2 between models are also commonly considered²⁴.

This approach, however, relates to the differences in the in-sample performance of competing models using different subsets of factors, whilst from a time series perspective, it is more relevant to evaluate the differences in the out-of-sample accuracy. Factor-augmented models are indeed adopted in finance with the primary aim of improving out-of-sample forecast accuracy. Therefore, in this setting the goal is to explain the marginal contribution of each factor $f_{k,t}$ in terms of out-of-sample forecasting. The factors contributing most to the out-of-sample performance of a factor-augmented model are the most important. In this regard²³, recently developed performance-based Shapley values (PBSVs) which decompose the out-of-sample accuracy of a machine learning model. The PBSVs shed light on how each predictor increases or decreases the out-of-sample accuracy. In what follows, we apply this rationale to the case of nonlinear latent factors obtained with autoencoders.

To compute a forecast-based Shapley value, we need to introduce a rolling window procedure. The used relation for one-step ahead forecasts is

$$\hat{x}_{i,t+1} = \hat{\alpha} + \sum_{j=1}^p \hat{\beta}_j x_{i,t-j+1} + \sum_{j=1}^p \hat{\gamma}_j' \hat{\mathbf{f}}_{t-j+1}. \quad (6)$$

Assuming that T ($t = 1, \dots, T$) total observations are available, we consider an estimation window equal to $M < T$. The first M observations ($t = 1, \dots, M$) constitute the training dataset of the first window and the remaining $D = T - M$ observations represent the out-of-sample dimension. Given an initial period from $t = 1$ to $t = M$, we train an autoencoder with a given architecture and estimate K latent factors $\mathbf{f}_t$ as explained in the previous subsection. We then estimate the factor-augmented models as in (4) and obtain one-step ahead forecasts $\hat{x}_{M+1}$, according to (6). Following a rolling-window procedure, in the next step $t + 1$ we estimate the factors $\mathbf{f}_t$ with the autoencoder using observations from $t = 2$ to $t = M + 1$ and obtain the forecasts $\hat{x}_{M+2}$ with the factor-augmented model. The procedure continues until no new observation is available and $\hat{x}_T$. The result is a D -dimensional vector of sample forecasting errors with generic d -th components given by $\hat{e}_d = (x_d - \hat{x}_d)$, with $d = M + 1, \dots, T$. To compute the Shapley value of the k -th factor as in (5) we consider the following expression

$$\xi_Q(\mathbf{f}_t) = E[(x_t - \hat{x}_t)^2 | F_{j,t} = f_{j,t}, \forall j \in Q], \quad (7)$$

which measures the change in the out-of-sample forecast accuracy, conditional on the factors in coalition Q that are used to obtain $\hat{x}_t$, when the predictor k is included in the conditioning information set. Therefore, we measure the Shapley value as the difference between the out-of-sample forecasting error of the factor-augmented model with and without the k -th factor included, weighted across all possible combinations of factors. In this setting, a negative value indicates that adding that k -th factor to the model reduces the overall prediction error in out-of-sample compared to the case where the factor is not included. Thus, using (5) and the definition provided in (7), a negative value of ϕ_k suggests that the k -th factor positively contributes to the predictive capability of the factor-augmented model when included. In other words, a negative Shapley value indicates that the factor is important and its inclusion improves the overall out-of-sample performance of the model. The more negative the Shapley value, the more important the factor.

We set the neural network to identify a specific number of latent variables, choosing $K = 5$. These latent factors, numbered from 1 to 5, explain the historical series of commodity returns. By setting $K = 5$, we ensure a comprehensive analysis while maintaining a balance between model complexity and interoperability.

Data

For the empirical analysis, we take into consideration the daily returns in dollars of $N = 22$ commodities, comprising 4424 observations, from October 4th, 2005, to November 18th, 2022. In particular, we consider log-returns of the i -th commodity, given by $x_{i,t} = \log(p_{i,t}/p_{i,t-1})$, with $p_{i,t}$ the price of the i -th commodity at time t . There are several categories into which commodities can be divided. For the empirical study, we select some of the most significant ones in the context of economics and finance³². These include commodities in the energy sector (Brent, Diesel, Gasoline, Natural Gas, and WTI); commodities in the metals sector (Aluminium, Copper, Gold, Nickel, Silver, and Zinc); and commodities in the agricultural sector (Coffee, Corn, Cotton, Lean Hogs, Live Cattle, Sugar, Soybeans, Soybean Meal, Soybean Oil, Sugar, HRW Wheat and Wheat).

The main statistics summary for the dataset is shown in Table 1. In terms of variability and shape, the daily return distributions are highly diverse. In fact, there are some notable variances in volatility even though all commodities exhibit nearly zero mean returns over the period under consideration.

Observe that the largest values of standard deviations are shown by the energy sector (WTI, Natural Gas and Gasoline), whilst smaller values are shown by gold—a classical safe haven—and live cattle. The commodity Lean Hogs is associated with the maximum value, whilst the commodity WTI has an unexpectedly small value at the minimum. This is because, on April 20th, 2020, the price closed at $-37.6\$$, which was the first time in history that the price was negative. Finally, a heavy-tailed distribution of returns is indicated by the strong kurtosis and the negative skewness of half of the commodities.

Figure 2 shows the time series of the returns on commodities; we highlight a few particular sub-periods with quite high levels of volatility. The more significant oscillations happen during periods of major economic crises, such as sovereign debt, subprime mortgages, and the recent COVID-19 pandemic.

As expected, commodities with low levels of oscillations include Gold and Silver. However, other metal commodities also exhibit fluctuation levels lower than, say, many agricultural commodities, where changes are more likely to occur without clear connections with periods of crisis. A significant exception is Nickel, which exhibited significant return volatility during the Russo-Ukrainian war along with natural gas and wheat.

As final remarks, note that almost all energy commodities (Brent, Diesel, Petrol, and WTI) are the ones where the pandemic impacted more. There are noticeable fluctuations around the mean values in every time series.

Results

As a premise, we adopt the BIC criterion for selecting the number of lags p in (4). Table 2 contains the results of the BIC algorithm obtained by the estimation of several linear factor-augmented models from lag $p = 1$ to lag $p = 5$. We find that the simple model with $p = 1$ with factors estimated using autoencoders provides the best fitting for all the considered commodities. Therefore, we keep the simple structure of $p = 1$ throughout the rolling-window experiment.

Commodity	Minimum	Maximum	Mean	SD	Kurtosis	Skewness
Aluminium	-10.77	9.75	0.02	1.49	6.08	-0.09
Brent	-24.40	21.02	0.03	2.33	13.62	-0.23
Coffee	-10.50	12.51	0.03	1.99	4.94	0.26
Copper	-10.87	12.35	0.03	1.75	6.95	0.03
Corn	-23.56	13.61	0.04	1.87	13.04	-0.61
Cotton	-23.88	11.10	0.03	1.83	12.63	-0.58
Diesel	-21.93	13.15	0.04	2.21	11.09	-0.43
Gasoline	-31.98	25.10	0.04	2.66	19.58	-0.36
Gold	-9.35	9.01	0.04	1.14	8.50	-0.21
HRW Wheat	-8.60	8.43	0.04	1.96	4.44	0.18
Lean Hogs	-20.91	26.66	0.03	2.35	28.49	0.34
Live Cattle	-14.48	7.25	0.02	1.13	15.84	-0.87
Natural Gas	-25.95	46.48	0.04	3.46	15.73	1.08
Nickel	-26.88	63.46	0.05	2.72	80.93	3.08
Silver	-17.75	12.97	0.05	2.08	8.92	-0.57
Soybean Meal	-18.55	10.87	0.04	1.85	13.95	-1.13
Soybean Oil	-8.86	7.79	0.04	1.52	5.70	-0.03
Soybeans	-10.50	6.69	0.03	1.52	7.75	-0.65
Sugar	-11.63	13.95	0.03	2.06	6.13	0.11
Wheat	-10.68	21.78	0.04	2.11	7.29	0.41
WTI	-305.97	37.66	-0.04	5.66	1987.80	-38.11
Zinc	-27.16	23.36	0.04	2.20	19.18	-0.24

Table 1. Descriptive statistics for the commodities dataset The table's columns indicate the minimum and maximum values, and the mean, standard deviation, kurtosis, and skewness.

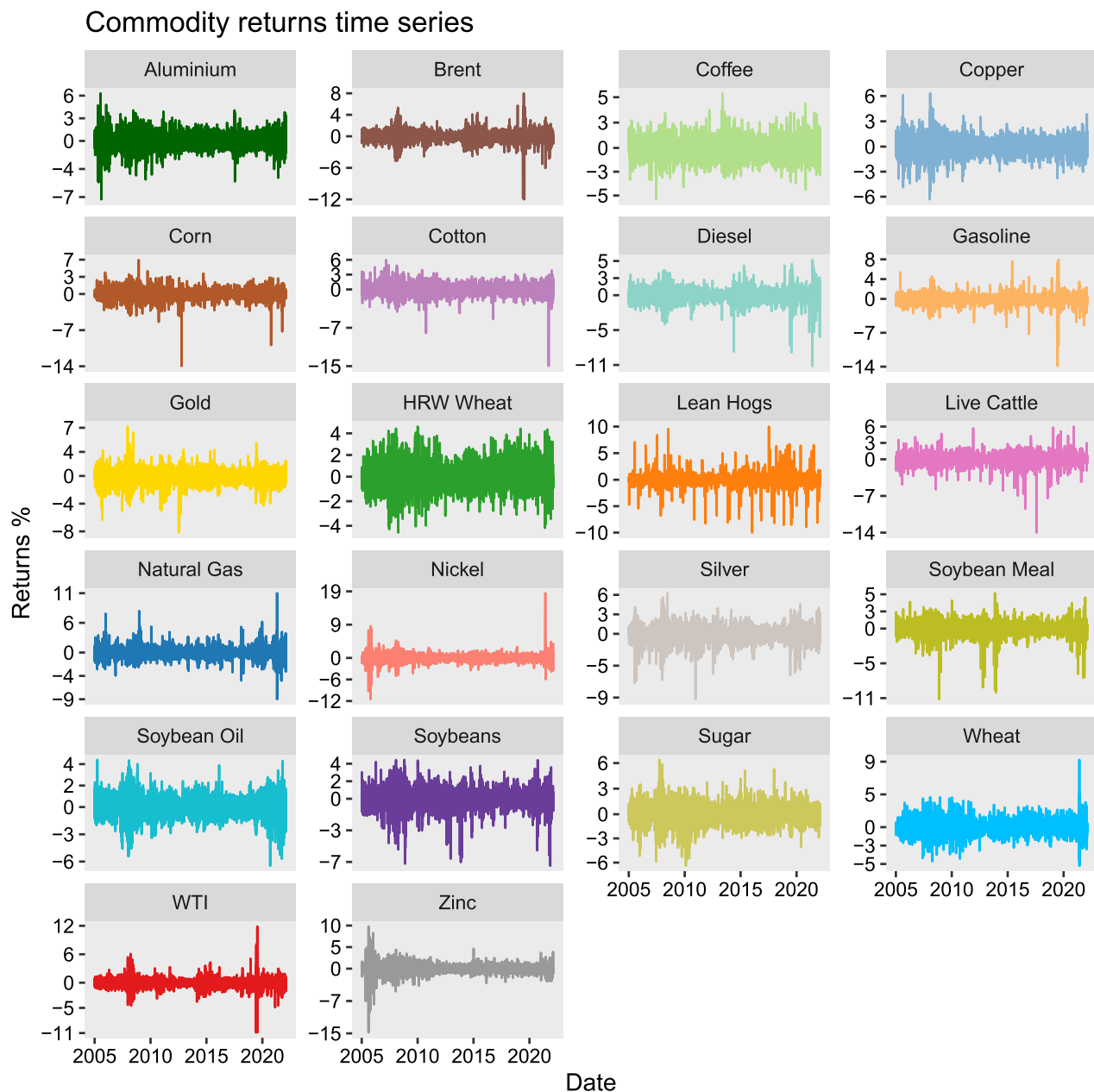


Figure 2. Temporal evolution of the commodity growth rates, computed using logarithmic price differentials and, subsequently, standardized for comparative analysis.

As already said above, the autoencoder enables us to reduce the dimensionality of the original data and identify the key factors influencing commodity prices. This approach allows us to better understand the relative importance of each latent factor in the context of commodity time series. It is important to note that the Shapley value represents the average contribution of a latent variable across all possible combinations of $N - 1$ latent variables, where the excluded variable is the one for which the Shapley value is being calculated. This method ensures a comprehensive assessment of each latent variable's marginal contribution to the explanation and prediction of commodity returns.

Initially, we assess the influence of each latent factor on the selected commodities sectors (energy, metals, and agriculture). Then, we analyze the contribution of each latent factor to individual commodities within these sectors. We consider the forecasting performance of the factors in the last month to assess their relevance using the procedure explained in Section 2. This detailed examination reveals how specific factors drive the returns of individual commodities, providing insights into the underlying dynamics of commodity markets.

In particular, we have organized our results to maximize the number of insights obtained. Figure 3 identifies the most relevant factors across our sample of commodities. This figure clearly shows that the latent variables Factor 1, Factor 2, and Factor 3 are able to explain and forecast the largest number of commodities. Specifically, these latent variables provide the highest marginal contributions, collectively being the most significant in five cases

Commodity	Number of lags				
	1	2	3	4	5
Natural Gas	-37.84	0.04	38.40	75.07	106.37
Gold	55.09	89.14	122.90	154.02	190.58
WTI	42.13	79.12	115.04	134.61	165.31
Brent	44.80	81.02	121.16	149.30	179.54
Soybeans	56.35	95.93	133.09	172.84	211.11
Corn	69.08	105.93	143.39	185.01	224.80
Copper	18.56	57.52	94.65	132.88	169.37
Silver	37.17	76.18	107.31	144.34	183.91
Live Cattle	53.38	84.29	118.46	156.16	195.01
Soybean Oil	40.28	78.72	113.46	147.60	185.26
Aluminium	11.71	46.29	80.00	116.29	150.28
Soybean Meal	59.87	98.28	135.38	175.19	214.37
Zinc	-45.43	-22.46	14.48	43.32	71.86
Diesel	-26.38	4.44	40.31	62.90	99.63
Nickel	-4.68	31.91	69.80	99.18	138.24
Wheat	51.31	87.03	127.41	165.61	202.27
Sugar	66.78	105.02	136.04	174.26	211.68
Gasoline	37.73	74.39	107.50	138.31	167.80
Coffee	25.54	60.39	95.72	131.62	170.94
Lean Hogs	52.37	88.45	120.58	160.92	200.77
HRW WHEAT	50.66	87.38	126.81	166.29	205.78
Cotton	-12.77	24.82	60.98	83.96	119.57

Table 2. Factor-augmented linear model (4): BIC values for the considered commodities and a different number of lags Results from $p = 1$ to $p = 5$. The lowest the value, the best the model.

each. Factor 1 is most significant for the commodities WTI, Brent, Soybean Oil, Wheat, and HRW Wheat. Factor 2 is most significant for Natural Gas, Corn, Copper, Aluminium, and Sugar. Factor 3 is most significant for Live Cattle, Nickel, Gasoline, Lean Hogs, and Cotton, whilst Factor 5 is the most relevant in four cases, namely Gold, Silver, Soybean Meal, and Coffee. Finally, Factor 4 is highly significant in three cases, Soybeans, Zinc, and Diesel.

Then, we provide an exploration of the average Shapley value of the five factors. In Fig. 4, Panel A presents the average Shapley value of all the commodities. In Panels B, C, and D, we conduct the same analysis but differentiating commodities by sector. The results are particularly intriguing. As they show, on average, the most relevant factors are consistently Factor 3 and Factor 5. This holds in almost all cases except for the energy sector, where Factor 1 emerges as the most important one. However, factors 3 and 5 still claim the 2nd and 3rd positions in this sector. These results are counter-intuitive considering our previous observation (see Fig. 3), where factors 1, 2, and 3 were found to be the best-performing for five commodities. This discrepancy might be explained by the specific characteristics of the energy sector compared to other sectors. Factor 1's dominance in the energy sector suggests that there are unique underlying factors driving commodity returns in this sector, which are different from those in the metals and agricultural sectors. Despite this, factors 3 and 5 consistently demonstrate their importance across all sectors, indicating their robust explanatory power. Such counter-intuitive findings challenge conventional wisdom and suggest that deeper investigation into the specific dynamics of each sector is warranted.

Finally, we examine the Shapley values from a disaggregated perspective, providing a comprehensive overview of the impact on individual commodities. Additionally, we represent their distribution using box plots, allowing us to extract some relevant information about the characteristics of the results' distribution. Fig. 5 represents Factor 1 for all commodities, whilst Figs. 6, 7, 8, and 9 provide the same information for factors 2, 3, 4, and 5, respectively. This approach enables us to gain insights into the individual contributions of each factor across commodities and understand the variability in their impacts. Box plots offer a visual representation of the distribution of Shapley values, allowing us to identify outliers, central tendency, and variability in the contributions of each factor. Delving deeper, we note that in Fig. 5, Factor 1 has the highest value for the commodities Sugar and Cotton, whilst it ranks third for WTI, despite it being the most prominent factor for WTI. For positive Shapley values, we can also mention Brent, Soybean Oil, Wheat, and HRW Wheat, where Factor 1 is the most relevant one. Here, the value suggests that the contribution of Factor 1 is marginal but more informative than other factors. Regarding the distribution, we want to highlight that the interquartile range appears to be larger compared to the others. In Fig. 6, we observe that Sugar, Cotton, Corn, Natural Gas, Aluminium, and Copper are the commodities where Factor 2 provides the highest explanatory power. In this case, only the commodity Cotton has the best score in terms of Shapley value, despite being placed outside of Fig. 3. Regarding the distribution, we note an outlier, specifically the Brent commodity. Observing Fig. 7, we find that Cotton and Sugar rank the highest, whilst Live Cattle, Nickel, Gasoline, and Lean Hogs are ranked lower in comparison, as reflected in Fig. 3. The Shapley value for Cotton in this case is the highest in absolute terms when considering all factors. On this, the

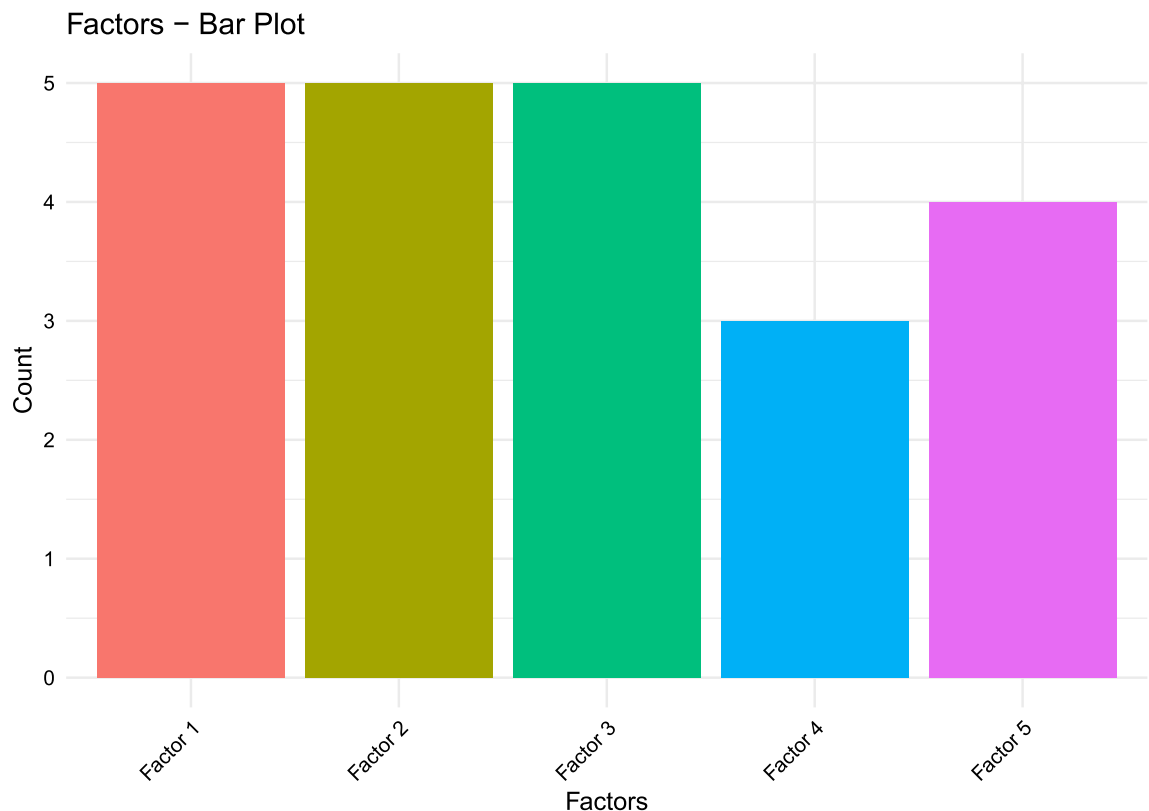


Figure 3. Number of times each factor predominates for the raw materials. Factor 1 is most significant for the commodities WTI, Brent, Soybean Oil, Wheat, HRW Wheat. Factor 2 is most significant for the commodities Natural Gas, Corn, Copper, Aluminium, Sugar. Factor 3 is most significant for Live Cattle, Nickel, Gasoline, Lean Hogs, Cotton. Factor 4 is most significant for Soybeans, Zinc, Diesel. Factor 5 for Gold, Silver, Soybean Meal, Coffee.

distribution identifies Cotton as an outlier. In Fig. 8, Factor 4 is most significant for Soybeans, Zinc, and Diesel. However, we also observe high Shapley values for Cotton and Sugar. Notably, Zinc has the highest value and is included in Fig. 3. Regarding the distribution, we observe a single outlier, which is again Brent.

Finally, Fig. 9 reports the Shapley value and the corresponding distribution of Factor 5. We see that the commodities with the lowest Shapley value are Cotton and Sugar, as we observe in other cases. We also observe that the value for Natural Gas is zero. The commodities for which Factor 5 is the predominant latent variable present negative Shapley value. Silver, for instance, ranks third in the predominance of Factor 5 after Sugar and Cotton, whilst Gold ranks fifth. The distribution does not exhibit any outliers; however, the interquartile range is noteworthy. The box plot shows that the median is slightly above zero, but the larger part of the box is composed of negative values, indicating that most of the Shapley values are negative.

Conclusions

This paper explores the explainability in the context of nonlinear factor models, which are grounded on neural networks and represent the machine learning counterpart of the principal component analysis. Due to their flexibility and good performance, autoencoders have been recently used for estimating nonlinear factor models in finance. However, the main weakness of autoencoders is that the results are less explainable than those obtained with the PCA.

In our analysis, we apply the concept of Shapley Value to evaluate the importance of individual k -th factor in a predictive model and, particularly, to improve the explainability of autoencoder factors. In particular, we measure the Shapley value as the difference between the out-of-sample forecasting error of the factor-augmented model with and without the k -th factor included, weighted across all possible combinations of factors.

We set the number of factors K to 5 for our analysis but acknowledged that it could be chosen differently depending on the problem's needs. We compute the Shapley values by comparing the model's prediction error when a single factor is included or excluded from the prediction, considering all possible combinations of factors, weighted according to the Shapley Value concept.

The methodological proposal is applied to a relevant set of the daily returns of 22 commodities, suitably divided by sectors. Results illustrate the heterogeneity of the explainability of the latent factors in the different contexts. Factors 1, 2 and 3 are predominant in explaining a large part of the commodities. Differently, taking the values of the Shapley values over all the commodities leads to a predominance of factors 3 and 5. When dealing with the individual commodities, we can grasp the relevance of each factor for all the commodities. On this, Sugar and Cotton are the most explainable commodities, whilst Brent is the least explained one. In the analysis

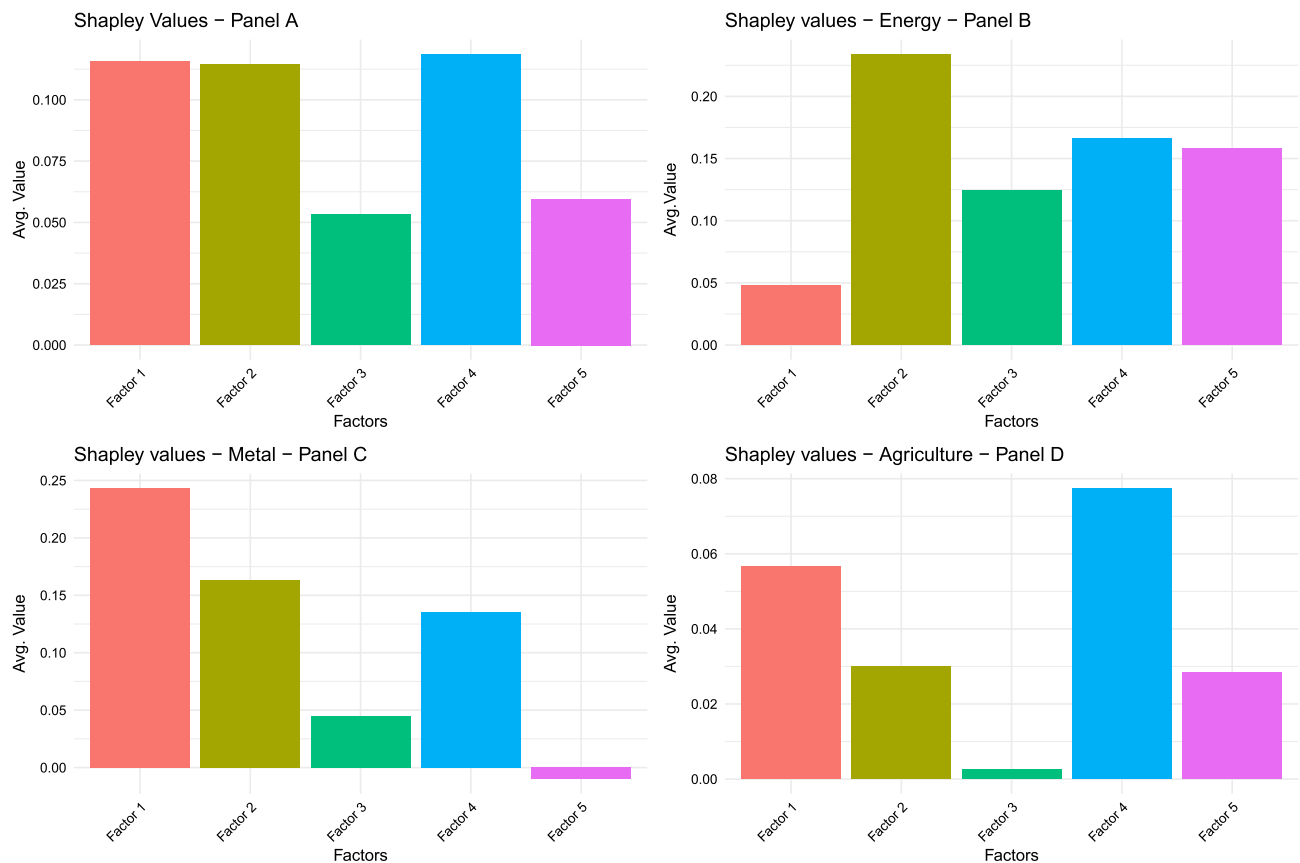


Figure 4. Panel A shows the average Shapley value per factor across all commodities, whilst Panel B displays the average factor values for the energy sector, Panel C for the metal sector, and Panel D for the agriculture sector.

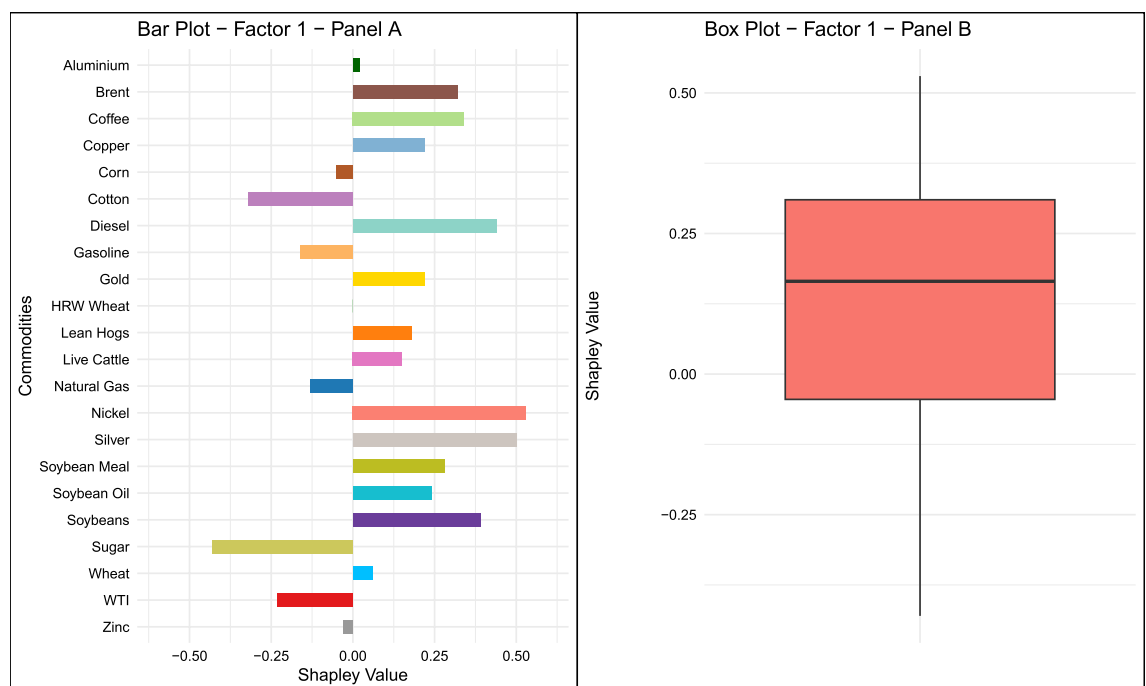


Figure 5. Panel A lists the Shapley value for all commodities, whilst panel B displays its distribution through the box plot. In the box plot of panel B, the box represents the interquartile range (IQR), which is the range between the first quartile (Q1) and the third quartile (Q3) of the data distribution. The whiskers extend up to 1.5 times the IQR from the top and bottom of the box, and outliers are represented by points beyond the whiskers.

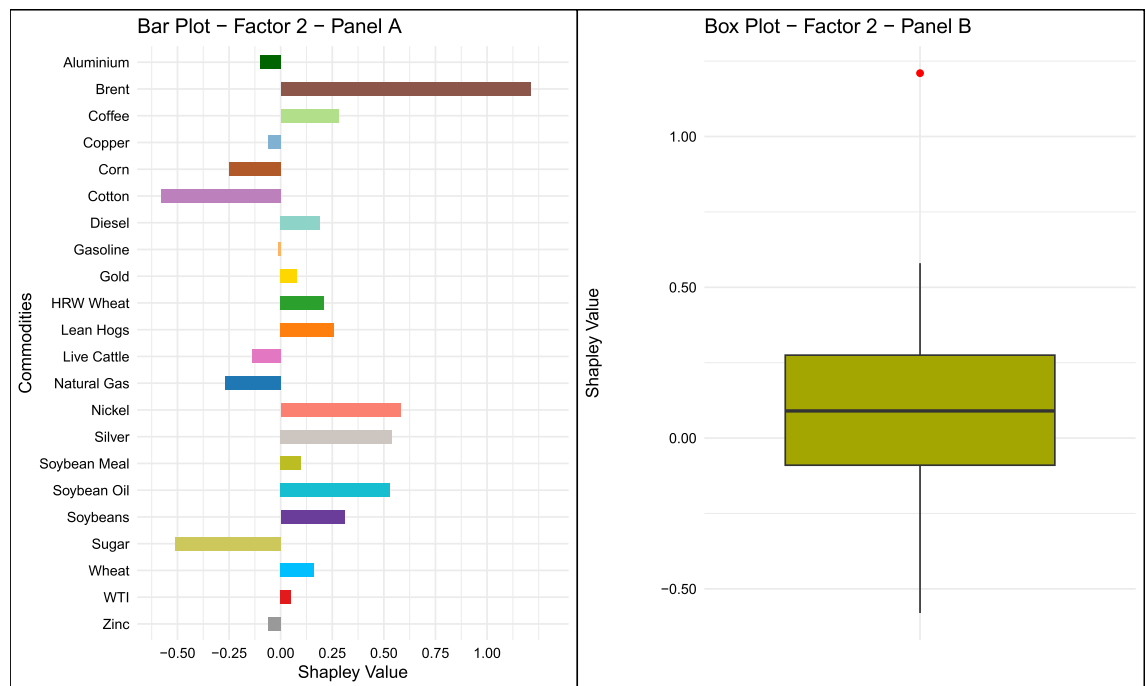


Figure 6. Case of Factor 2. See the caption of Figure 5; additionally, here we notice a red point representing an outlier.

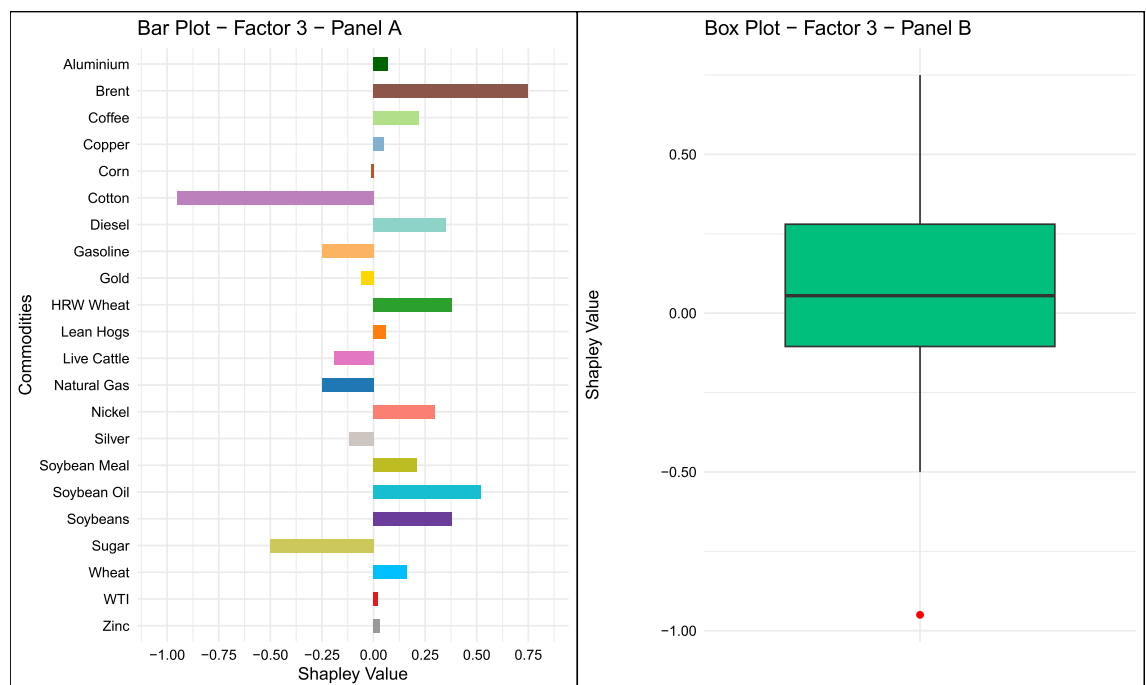


Figure 7. Case of Factor 3. Panel A reports the Shapley value for all commodities, whilst panel B displays the distribution of its value.

at a sector level, we have the predominance of Factor 1 for energy, Factor 5 for metal—and, interestingly, Factor 5 is the unique negative one for this sector—and Factor 3 for agricultural commodities.

In the end, some future research directions can be highlighted. First, we note that the forecast-based Shapely value procedure proposed in the paper assumes that the number of factors K is fixed. Therefore, the researchers can select an arbitrary number of factors (not necessarily $K = 5$) in their empirical studies following the same methodology. However, while the selection of the number of factors in the context of principal component analysis is well-developed³³, e.g. see, less is known for autoencoders. The choice of K therefore remains open for further consideration and investigation, as it could affect the results. Future work may involve exploring how the choice of K influences the results and whether K should be adapted based on the specific context.

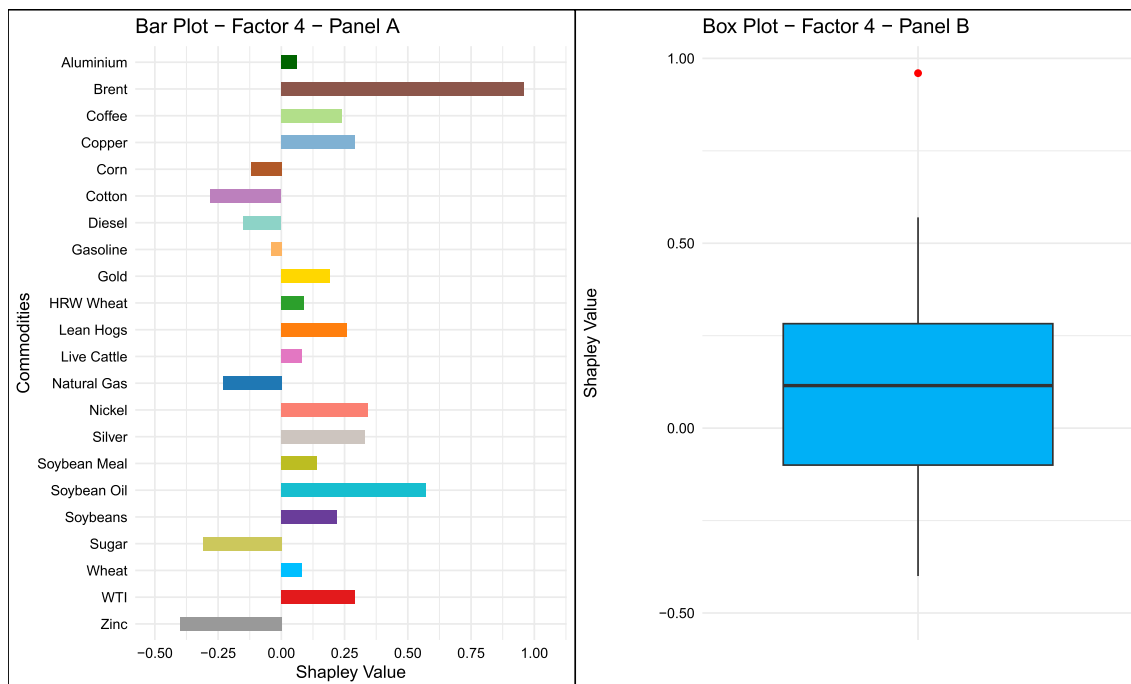


Figure 8. Shapley value and box plot for Factor 4. See the captions of Figs. 5, 6, and 7.

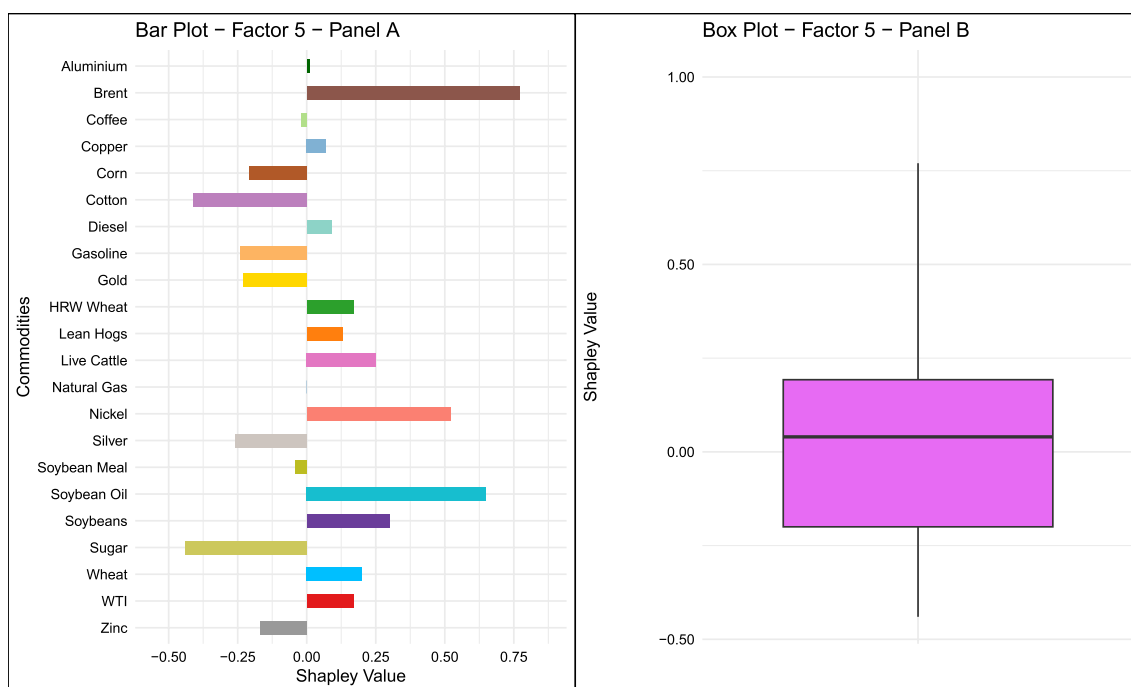


Figure 9. Case of Factor 5. See the captions of Figs. 5, 6, 7, and 8.

Another aspect deserving of future research is the impact of the out-of-sample dimension adopted in the computational procedure for the Shapley value calculation. Specifically considering an out-of-sample dimension would modify the results. This applies to any forecast-based methodology since the forecast losses—which are the main ingredient of the Shapley value procedure adopted in the paper—can assume different values. In particular, for our empirical analysis with commodities, we consider one month of daily observations for the out-of-sample testing, which is enough for our purposes.

Indeed, the proposed out-of-sample experiments provides a clear validation of the methodological Shapley value-based proposal. Moreover, we stress that the computational complexity of the problem does not allow to deal with particularly large windows, since in our context the computation of the Shapley value involves the

analysis of the partitions of a set for each time window—and the number of the partitions of a discrete set with cardinality n is the huge Bell number $B(n)$. Interestingly, one can deal with the optimal size of such an out-of-sample dimension in our forecasting procedures. This challenging task is left for future research.

Data availability

The datasets generated during and analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

R. C., A. I., R. M. and S. S. designed the study, performed the experiments, analyzed the data, and wrote the manuscript. All authors participated in the discussions of the results and manuscript revisions. All authors agreed with the content of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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